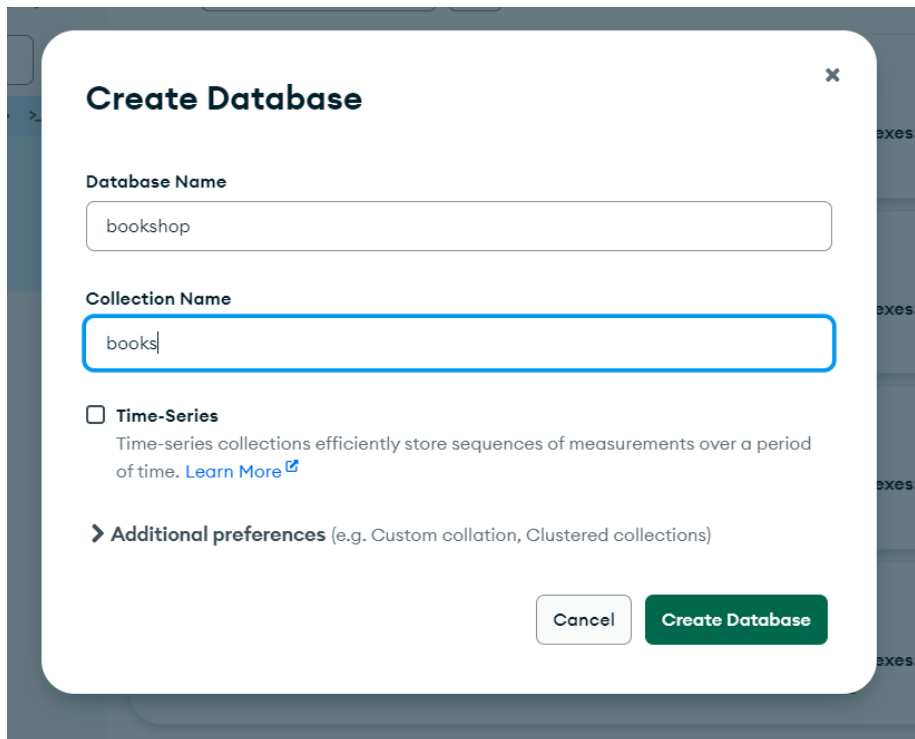


MongoDB Exercise

1. create a database bookshop

create a collection books

A screenshot of the MongoDB 'Create Database' dialog box. The dialog has a title bar with a close button (X). It contains two input fields: 'Database Name' with the text 'bookshop' and 'Collection Name' with the text 'books'. Below these fields is a checkbox labeled 'Time-Series' which is unchecked. To the right of the checkbox is a description: 'Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)'. Below this is a section titled 'Additional preferences (e.g. Custom collation, Clustered collections)' with a right-pointing arrow. At the bottom right are two buttons: 'Cancel' and 'Create Database'.

2. Insert the following data

```
{  
  "title": "The Hobbit",  
  "author": "J.R.R. Tolkien",  
  "published_year": 1937,  
  "genres": ["Fantasy", "Adventure"],  
  "pages": 310,  
  "available": true,  
  "rating": 4.8
```

```
}
```

```
{
```

```
  "title": "1984",
```

```
  "author": "George Orwell",
```

```
  "published_year": 1949,
```

```
  "genres": ["Dystopian", "Political Fiction"],
```

```
  "pages": 328,
```

```
  "available": true,
```

```
  "rating": 4.7
```

```
},
```

```
{
```

```
  "title": "To Kill a Mockingbird",
```

```
  "author": "Harper Lee",
```

```
  "published_year": 1960,
```

```
  "genres": ["Southern Gothic", "Drama"],
```

```
  "pages": 281,
```

```
  "available": false,
```

```
  "rating": 4.6
```

```
},
```

```
{
```

```
  "title": "The Great Gatsby",
```

```
  "author": "F. Scott Fitzgerald",
```

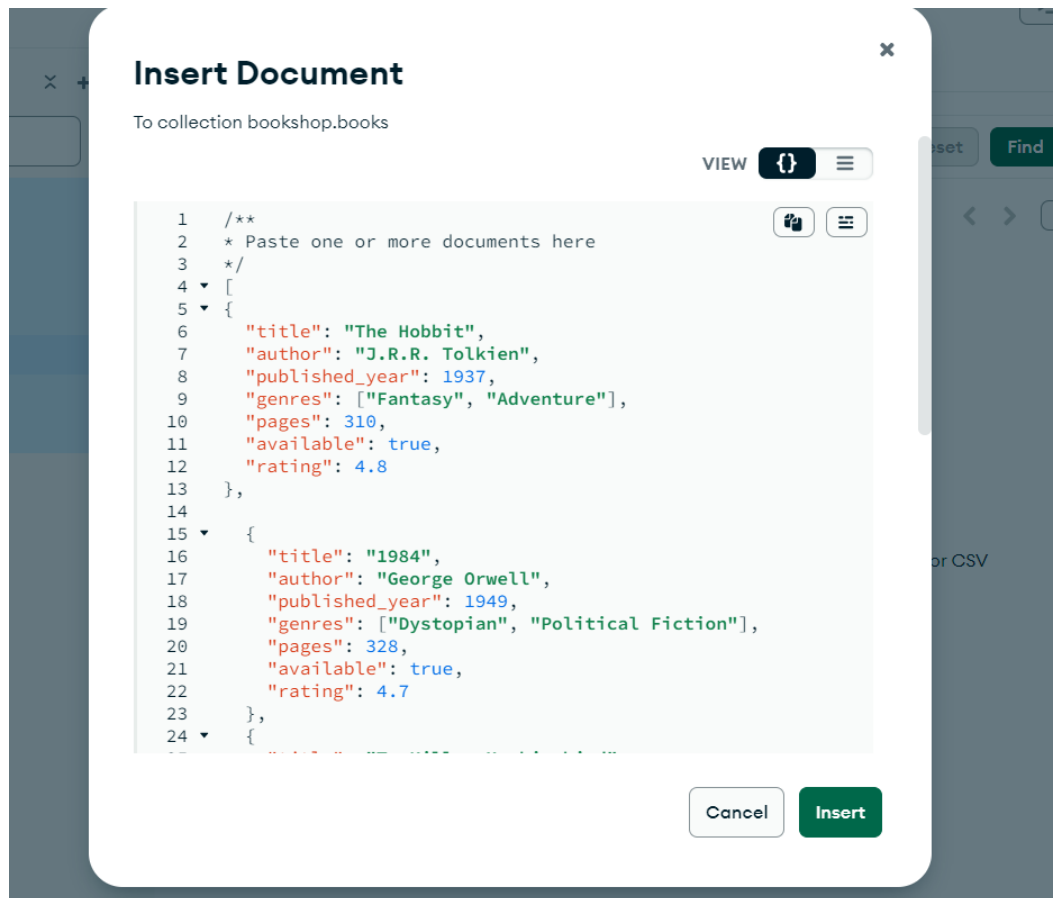
```
  "published_year": 1925,
```

```
  "genres": ["Tragedy"],
```

```
  "pages": 180,
```

```
  "available": true,
```

```
"rating": 4.4
},
{
  "title": "Brave New World",
  "author": "Aldous Huxley",
  "published_year": 1932,
  "genres": ["Science Fiction", "Dystopian"],
  "pages": 311,
  "available": true,
  "rating": 4.2
}
```



Insert Document

To collection bookshop.books

```
24 {
25   "title": "To Kill a Mockingbird",
26   "author": "Harper Lee",
27   "published_year": 1960,
28   "genres": ["Southern Gothic", "Drama"],
29   "pages": 281,
30   "available": false,
31   "rating": 4.6
32 },
33 {
34   "title": "The Great Gatsby",
35   "author": "F. Scott Fitzgerald",
36   "published_year": 1925,
37   "genres": ["Tragedy"],
38   "pages": 180,
39   "available": true,
40   "rating": 4.4
41 },
42 {
43   "title": "Brave New World",
44   "author": "Aldous Huxley",
45   "published_year": 1932,
46   "genres": ["Science Fiction", "Dystopian"],
47   "pages": 311,
48   "available": true,
49   "rating": 4.2
50 }
```

Cancel

Insert

Inserted Data:

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

ADD DATA

EXPORT DATA

UPDATE

DELETE

25

1 - 5 of 5

↺

↻

↷

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☰

```
_id: ObjectId('681872cbab822f3df7a068f6')
title: "The Hobbit"
author: "J.R.R. Tolkien"
published_year: 1937
genres: Array (2)
pages: 310
available: true
rating: 4.8
```

```
_id: ObjectId('681872cbab822f3df7a068f7')
title: "1984"
author: "George Orwell"
published_year: 1949
genres: Array (2)
pages: 328
available: true
rating: 4.7
```

```
_id: ObjectId('681872cbab822f3df7a068f8')
title: "To Kill a Mockingbird"
author: "Harper Lee"
published_year: 1960
genres: Array (2)
pages: 281
available: false
rating: 4.6
```

Type a query: { field: 'value' } or [Generate query](#) ✨

[EXPLAIN](#)
[RESET](#)
[FIND](#)
[</>](#)

[ADD DATA](#)
[EXPORT DATA](#)
[UPDATE](#)
[DELETE](#)

25 1 - 5 of 5

```

_id: ObjectId('681872cbab822f3df7a068f8')
title: "To Kill a Mockingbird"
author: "Harper Lee"
published_year: 1960
genres: Array (2)
pages: 281
available: false
rating: 4.6

```

```

_id: ObjectId('681872cbab822f3df7a068f9')
title: "The Great Gatsby"
author: "F. Scott Fitzgerald"
published_year: 1925
genres: Array (1)
pages: 180
available: true
rating: 4.4

```

```

_id: ObjectId('681872cbab822f3df7a068fa')
title: "Brave New World"
author: "Aldous Huxley"
published_year: 1932
genres: Array (2)
pages: 311
available: true
rating: 4.2

```

3. Find all books

db.books.find()

```

>_MONGOSH
> db.bookshop.find()
< {
  _id: ObjectId('6818723ec3fd4e6bc8051791'),
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  published_year: 1937,
  genres: [
    'Fantasy',
    'Adventure'
  ],
  pages: 310,
  available: true,
  rating: 4.8
}
{
  _id: ObjectId('6818723ec3fd4e6bc8051792'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  pages: 328,
  available: true,
  rating: 4.7
}

```

```
>_MONGOSH
{
  _id: ObjectId('6818723ec3fd4e6bc8051793'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  published_year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  pages: 281,
  available: false,
  rating: 4.6
}
{
  _id: ObjectId('6818723ec3fd4e6bc8051794'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  published_year: 1925,
  genres: [
    'Tragedy'
  ],
  pages: 180,
  available: true,
  rating: 4.4
}
{
  _id: ObjectId('6818723ec3fd4e6bc8051795'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  published_year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  pages: 311,
  available: true,
  rating: 4.2
}
bookshop>
```

4.Find all books published after 1950

```
db.books.find({ published_year: { $gt: 1950 } })
```

```
> db.books.find({ published_year: { $gt: 1950 } })
< {
  _id: ObjectId('681872cbab822f3df7a068f8'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  published_year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  pages: 281,
  available: false,
  rating: 4.6
}
bookshop>
```

5. Find the available books

`db.books.find({ available: true })`

```
> db.books.find({ available: true })
< {
  _id: ObjectId('681872cbab822f3df7a068f6'),
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  published_year: 1937,
  genres: [
    'Fantasy',
    'Adventure'
  ],
  pages: 310,
  available: true,
  rating: 4.8
}
{
  _id: ObjectId('681872cbab822f3df7a068f7'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  pages: 328,
  available: true,
  rating: 4.7
}
```

```
{
  _id: ObjectId('681872cbab822f3df7a068f9'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  published_year: 1925,
  genres: [
    'Tragedy'
  ],
  pages: 180,
  available: true,
  rating: 4.4
}
{
  _id: ObjectId('681872cbab822f3df7a068fa'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  published_year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  pages: 311,
  available: true,
  rating: 4.2
}
bookshop >
```

6. Find all books that belong to the "Dystopian" genre

```
db.books.find({ genres: "Dystopian" })
```

```
> db.books.find({ genres: "Dystopian" })
< {
  _id: ObjectId('681872cbab822f3df7a068f7'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  pages: 328,
  available: true,
  rating: 4.7
}
{
  _id: ObjectId('681872cbab822f3df7a068fa'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  published_year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  pages: 311,
  available: true,
  rating: 4.2
}
bookshop>
```

7. Find the books published after 1960 and had rating more than 4.0

```
db.books.find({ published_year: { $gt: 1960 }, rating: { $gt: 4.0 } })
```

```
<
> db.books.find({ published_year: { $gt: 1960 }, rating: { $gt: 4.0 } })
<
bookshop>
```

8. Project only title and author fields

```
db.books.find({}, { title: 1, author: 1, _id: 0 })
```



```

> db.books.find({}, { title: 1, author: 1, _id: 0 })
< {
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien'
}
{
  title: '1984',
  author: 'George Orwell'
}
{
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee'
}
{
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald'
}
{
  title: 'Brave New World',
  author: 'Aldous Huxley'
}
}
bookshop>

```

9. Update the book "1984" to set available: false

```

db.books.updateOne(
  { title: "1984" },
  { $set: { available: false } }
)

```

```

> db.books.updateOne(
  { title: "1984" },
  { $set: { available: false } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
bookshop>

```

```

_id: ObjectId('681872cbab822f3df7a068f7')
title: "1984"
author: "George Orwell"
published_year: 1949
▶ genres: Array (2)
pages: 328
available: false
rating: 4.7

```

10. Increase the rating of "Brave New World" by +0.1

```
db.books.updateOne(  
  { title: "Brave New World" },  
  { $inc: { rating: 0.1 } }  
)
```

```
> db.books.updateOne(  
  { title: "Brave New World" },  
  { $inc: { rating: 0.1 } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
bookshop>
```

```
▶ _id: ObjectId('681872cbab822f3df7a068fa')  
  title: "Brave New World"  
  author: "Aldous Huxley"  
  published_year: 1932  
  ▶ genres: Array (2)  
    pages: 311  
    available: true  
    rating: 4.3
```

11. Add a new genre "Classic" to "The Great Gatsby"

```
db.books.updateOne(  
  { title: "The Great Gatsby" },  
  { $addToSet: { genres: "Classic" } }  
)
```

```
> db.books.updateOne(  
  { title: "The Great Gatsby" },  
  { $addToSet: { genres: "Classic" } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
bookshop>
```

```
▶ {
  _id: ObjectId('681872cbab822f3df7a068f9')
  title: "The Great Gatsby"
  author: "F. Scott Fitzgerald"
  published_year: 1925
  genres: Array (2)
    0: "Tragedy"
    1: "Classic"
  pages: 180
  available: true
  rating: 4.4
}
```

12. Sort books by published year ascending

```
db.books.find().sort({ published_year: 1 })
```

```
>_MONGOSH
> db.books.find().sort({ published_year: 1 })
< {
  _id: ObjectId('681872cbab822f3df7a068f9'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  published_year: 1925,
  genres: [
    'Tragedy',
    'Classic'
  ],
  pages: 180,
  available: true,
  rating: 4.4
}
{
  _id: ObjectId('681872cbab822f3df7a068fa'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  published_year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  pages: 311,
  available: true,
  rating: 4.3
}
```

```

>_MONGOSH
{
  _id: ObjectId('681872cbab822f3df7a068f6'),
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  published_year: 1937,
  genres: [
    'Fantasy',
    'Adventure'
  ],
  pages: 310,
  available: true,
  rating: 4.8
}
{
  _id: ObjectId('681872cbab822f3df7a068f7'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  pages: 328,
  available: false,
  rating: 4.7
}

{
  _id: ObjectId('681872cbab822f3df7a068f8'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  published_year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  pages: 281,
  available: false,
  rating: 4.6
}
bookshop>

```

13. Sort books by rating descending

```
db.books.find().sort({ rating: -1 })
```

```

>_MONGOSH
> db.books.find().sort({ rating: -1 })
< {
  _id: ObjectId('681872cbab822f3df7a068f6'),
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  published_year: 1937,
  genres: [
    'Fantasy',
    'Adventure'
  ],
  pages: 310,
  available: true,
  rating: 4.8
}
{
  _id: ObjectId('681872cbab822f3df7a068f7'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  pages: 328,
  available: false,
  rating: 4.7
}

```

```

>_MONGOSH
{
  _id: ObjectId('681872cbab822f3df7a068f8'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  published_year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  pages: 281,
  available: false,
  rating: 4.6
}
{
  _id: ObjectId('681872cbab822f3df7a068f9'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  published_year: 1925,
  genres: [
    'Tragedy',
    'Classic'
  ],
  pages: 180,
  available: true,
  rating: 4.4
}

```

```

{
  _id: ObjectId('681872cbab822f3df7a068fa'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  published_year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  pages: 311,
  available: true,
  rating: 4.3
}
bookshop>

```

14. Delete all books with a rating lower than 4.5

```
db.books.deleteMany({ rating: { $lt: 4.5 } })
```

```
> db.books.deleteMany({ rating: { $lt: 4.5 } })
< {
  acknowledged: true,
  deletedCount: 2
}
bookshop>
```

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Optimize

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 3 of 3

```
_id: ObjectId('681872cbab822f3df7a068f6')
title: "The Hobbit"
author: "J.R.R. Tolkien"
published_year: 1937
genres: Array (2)
pages: 310
available: true
rating: 4.8
```

```
_id: ObjectId('681872cbab822f3df7a068f7')
title: "1984"
author: "George Orwell"
published_year: 1949
genres: Array (2)
pages: 328
available: false
rating: 4.7
```

```
_id: ObjectId('681872cbab822f3df7a068f8')
title: "To Kill a Mockingbird"
author: "Harper Lee"
published_year: 1960
genres: Array (2)
pages: 281
available: false
rating: 4.6
```